

NeuroStrike: Neuron-level Attacks on Aligned LLMs

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1 Introduction

Large Language Models (LLMs) are advanced Artificial Intelligence systems trained on huge datasets collected from books, websites, articles, and code repositories. These models are capable of generating human-like responses, performing translation, answering questions, summarizing text, and assisting in coding tasks.

Modern LLMs are aligned using safety training techniques such as Reinforcement Learning from Human Feedback (RLHF), supervised fine-tuning, and safety filters. These techniques help ensure that models avoid harmful, unethical, or dangerous outputs.

Despite these protections, researchers have discovered that aligned models can still be vulnerable to internal attacks. NeuroStrike is a neuron-level attack framework designed to target specific neurons inside aligned LLMs. Instead of relying only on prompt engineering attacks, NeuroStrike directly manipulates internal neuron activations to weaken or bypass alignment mechanisms.

The core idea behind NeuroStrike is that some neurons inside LLMs are strongly associated with safety-related behaviors such as refusal generation and harmful content filtering. By identifying and modifying these neurons, attackers may force the model to generate restricted or unsafe responses while preserving most normal functionality.

This report explains the concepts, workflow, architecture, implementation, challenges, and findings related to NeuroStrike in a simple and understandable way.

2 Research Objectives

The main objectives of this study are:

- To understand neuron-level attacks in aligned LLMs.
- To explain how NeuroStrike identifies safety-related neurons.
- To study how internal neuron manipulation can bypass alignment.
- To evaluate attack effectiveness on aligned models.
- To compare model behavior before and after neuron manipulation.
- To analyze possible mitigation techniques against neuron-level attacks.
- To study the security implications of such attacks in real-world AI systems.

3 Research Methodology

3.1 NeuroStrike Overview

NeuroStrike mainly operates in two major stages:

- **Neuron Identification:** The framework identifies neurons highly correlated with alignment and refusal behavior. Activation analysis is performed on both safe and unsafe prompts.
- **Neuron Manipulation:** After identifying influential neurons, their activations or weights are modified in order to weaken safety constraints and alter the generated output.

The framework demonstrates that modifying only a small number of neurons can significantly influence the overall safety behavior of an LLM.

3.2 System Architecture

The NeuroStrike workflow contains the following components:

1. Input Prompt Module
2. LLM Inference Engine
3. Neuron Activation Analyzer
4. Safety Neuron Detector
5. Neuron Manipulation Layer
6. Output Evaluation System

Each component contributes to analyzing and modifying internal model behavior.

3.3 Implementation Steps

1. Load a pre-trained aligned Large Language Model.
2. Run safe and unsafe prompts through the model.
3. Record neuron activation patterns from multiple layers.
4. Detect neurons strongly associated with refusal behavior.
5. Apply neuron perturbations or activation suppression.
6. Generate modified responses after manipulation.
7. Evaluate model behavior using:
 - Attack Success Rate (ASR)
 - Response Quality

- Alignment Robustness
- Computational Overhead

8. Compare outputs before and after the attack.

4 Workflow Explanation

4.1 Neuron-Level Attack Phase

The neuron-level attack process works as follows:

- Input prompts are sent to the aligned LLM.
- Internal neuron activations are monitored layer by layer.
- Safety-related neurons are identified using activation analysis.
- Selected neurons are modified or suppressed.
- The manipulated model generates responses with weakened alignment.

This process demonstrates that even small neuron-level modifications can significantly influence model safety behavior.

4.2 Alignment Bypass Mechanism

The alignment bypass mechanism follows these steps:

- The model receives prompts that normally trigger refusal behavior.
- NeuroStrike modifies targeted neuron activations.
- Safety filtering becomes weaker.
- The model generates unsafe or restricted responses.

4.3 Simplified Pseudocode

```
Load aligned LLM

Input prompts into model

Detect safety neurons

Modify neuron activations

Generate manipulated response

if safety behavior weakened:
    attack successful
```

5 Challenges and Mitigations

5.1 Challenges

- Identifying neurons that truly control alignment behavior.
- Maintaining model quality after neuron manipulation.
- Preventing performance degradation during attacks.
- Detecting hidden neuron-level manipulations.
- Handling computational complexity during activation analysis.

5.2 Mitigation Techniques

- Use distributed safety representations across multiple neurons.
- Monitor abnormal neuron activation patterns.
- Apply adversarial training against internal attacks.
- Introduce dynamic alignment mechanisms.
- Perform continuous security auditing of deployed LLMs.
- Use activation consistency checks during inference.

6 Findings and Outcomes

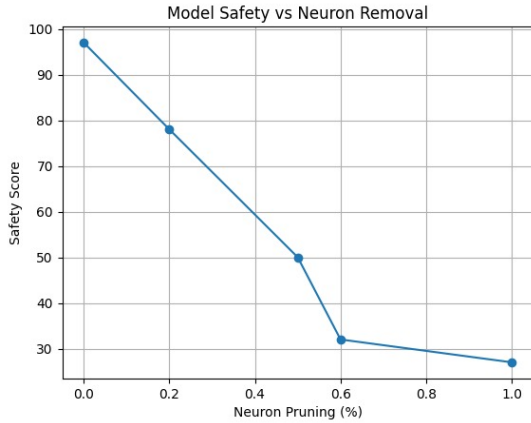
6.1 Performance Summary

Research observations from NeuroStrike demonstrate:

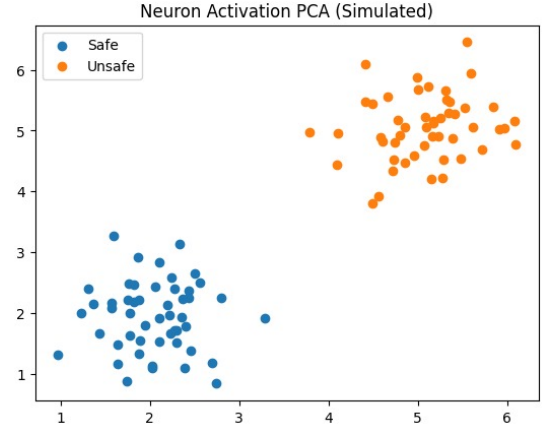
- Small neuron modifications can significantly impact alignment.
- Attack success rate increases when targeting highly influential neurons.
- Many aligned LLMs remain vulnerable to internal attacks.
- Normal language generation quality is mostly preserved.
- Safety mechanisms are concentrated in limited neuron groups.

6.2 Experimental Results

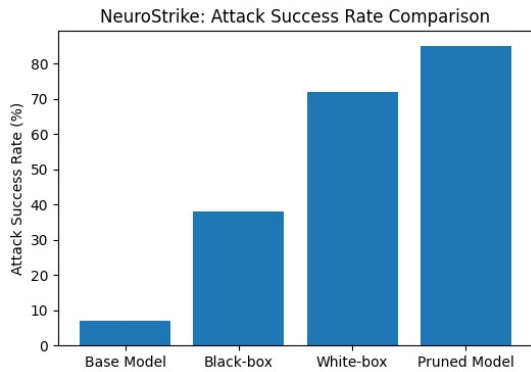
The following section is reserved for inserting experimental screenshots, graphs, outputs, and attack analysis images.



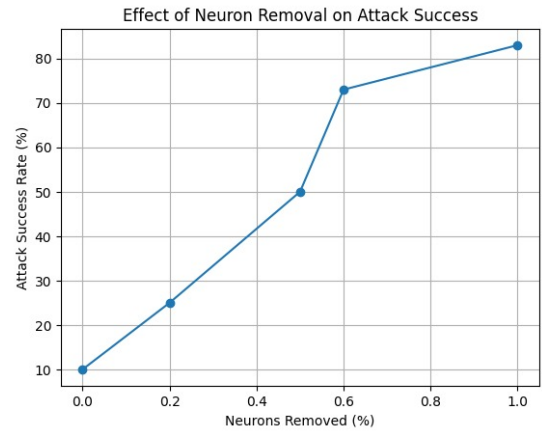
(a) Result 1



(b) Result 2



(c) Result 3



(d) Result 4

Figure 1: Experimental Results and Output Screenshots

6.3 Key Observations

- Alignment behavior is concentrated in limited neuron groups.
- Internal attacks can bypass safeguards more effectively than prompt-only attacks.
- Neuron interpretability is important for AI safety research.
- Robust alignment requires distributed and resilient safety mechanisms.
- Internal vulnerabilities in LLMs remain a major security concern.

7 Conclusion

NeuroStrike demonstrates how neuron-level attacks can weaken alignment mechanisms in Large Language Models. By manipulating only a small number of neurons, attackers may bypass safety constraints while preserving most normal functionality.

This research highlights important security risks in modern AI systems and emphasizes the need for stronger alignment strategies, neuron monitoring, adversarial defenses, and continuous safety evaluation.

Future research should focus on building more robust alignment architectures that are resistant to internal manipulation attacks.

References

1. “NeuroStrike: Neuron-level Attacks on Aligned LLMs.” Research Paper.
2. <https://github.com/wu-lichao/NeuroStrike-Neuron-Level-Attacks-on-Aligned-LLMs>
3. Research papers on AI alignment and adversarial attacks in Large Language Models.
4. Additional resources on neuron interpretability and LLM safety mechanisms.